

# Prompt Minimization: Reducing Input Redundancy Without Sacrificing Output Fidelity

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## Abstract

Despite the growing capabilities of large language models (LLMs), prompt design remains largely heuristic and ad hoc. This project will explore *prompt minimization*, the process of reducing prompts to their smallest, most information-dense form while preserving output fidelity. Practically, shorter prompts reduce computational overhead and inference latency (Vaswani et al., 2017), especially when large contexts, such as entire documents or codebases, are included unnecessarily. Further, longer prompts can damage LLM reasoning and accuracy (Levy et al., 2024). Theoretically, the existence of multiple prompts yielding equivalent outputs suggests a high degree of redundancy in the input space, raising fundamental questions about what information is essential to elicit specific model behaviors. We propose three variant frameworks to identify and evaluate minimal prompts and demonstrate that minimal prompts often produce outputs comparable to those of their longer counterparts. These findings suggest new directions for efficient prompt engineering and deepen our understanding of input compression in LLMs.

## 1 Introduction

Prompting for large language models (LLMs) is often redundant: many different prompts can elicit the same or near-identical outputs, yet practitioners routinely paste long documents or sprawling instructions “just in case.” These long inputs introduce additional latency and computational overhead (Vaswani et al., 2017), potentially resulting in substantial performance penalties for both response time and throughput. This contributes to the LLM speed and size bottlenecks in deployment, especially at scale, limiting the practical use of

LLMs.<sup>1</sup>

Further, long inputs can have damaging impacts on reasoning and accuracy. Across multiple models, lengthier prompts degrade accuracy, even when the longer prompt is just a duplication of the shorter prompt (Levy et al., 2024). Longer context consistently decreases accuracy for multiple models, even if the model retrieves relevant information perfectly (Du et al., 2025). For instance, large numbers of whitespace or masked tokens, which minimally distract models, decrease accuracy, signifying an intrinsic weakness in LLMs to longer inputs. In practice, longer prompts are likely to include irrelevant information, which can distract LLMs and confuse them. Even with mitigation strategies such as chain-of-thought and self-consistency, noisy prompts consistently and significantly decrease performance (Shi et al., 2023; Wang et al., 2024; Wu et al., 2024; Jiang et al., 2025). These findings highlight LLMs’ high susceptibility to distraction and their limited ability to distinguish relevant from irrelevant information.

Lastly, LLMs lack explainability, or the ability to explain how they arrived at their outputs (Doshi-Velez and Kim, 2017). This has significant implications for trustworthiness. LLMs are prone to hallucinations and may inherit bias from their training process. Without explainability of how the model generated its output, users cannot be sure whether a model’s output can be trusted (Zhao et al., 2024). Additionally, it is often challenging to design an ideal prompt for an LLM because users cannot see or understand how the model interprets and uses it. Better explainability would enable users to understand LLMs’ ca-

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pabilities and limitations and to design improved prompts for their end goals. Identifying shorter prompts that yield the same results as longer prompts may clarify how models interpret prompts and which information they rely on most, guiding prompt engineers in developing ideal prompts.

We target the core question behind this ambiguity: *what is the minimal natural-language prompt that preserves a model’s output fidelity for a given task and model?* This matters for (i) **efficiency**—shorter inputs reduce latency, token cost, and context pressure; (ii) **accuracy**—isolating the truly relevant information mitigates LLMs’ inherent difficulty in processing longer prompts and reduces distraction; and (iii) **scientific understanding**—characterizing equivalence classes of prompts reveals how models map input information to behavior.

## 2 Related work

Previous work has shown that prompting with In Context Learning (ICL) can improve LLM performance on downstream tasks (Brown et al., 2020). Similarly, Lester et al. (2021) showed that prompt tuning, in which the model parameters are fixed, and the prompt is adjusted, can achieve performance comparable to fine-tuning. The effectiveness of prompting has led to the emergence of prompt engineering as a field of study that focuses on using text instructions to extend the capabilities of LLMs (Sahoo et al., 2025). Applications include using prompting to enable reasoning with Chain-of-Thought (CoT) variants as first introduced by Wei et al. (2023).

Another sub-field of prompt engineering deals with optimizing the prompt itself for a given task (Schulhoff et al., 2025). Early approaches to prompt optimization focused on gradient-based methods (Shin et al., 2020); however, due to a vast search space and large models, they become computationally infeasible and too costly to scale (Zhang et al., 2024). Later techniques include gradient-free meta prompting (Schulhoff et al., 2025) and genetic algorithms (Zhang et al., 2024) to tune the prompt, both of which have shown promising results.

Past studies on prompt minimization have split mainly into two approaches: pruning,

or directly removing less relevant tokens, and summarization, changing the prompt itself while maintaining overall semantic meaning (Chang et al., 2024). Several pruning methods split prompts into sections (e.g., instruction, demonstration, question, etc.) and minimize the size of individual sections (Jiang et al., 2023; Zhou et al., 2023). Another pruning strategy is to consider token-wise importance and remove tokens deemed less important (Jiang et al., 2023; Jung and Kim, 2024; Li et al., 2023; Ali et al., 2024). These techniques have been shown to minimize prompt length while retaining output fidelity successfully.

Summarization methods are relatively less explored. These methods generally rely on ICL to compress long text while preserving its meaning. Many of these summarization methods focus on compressing context in retrieval-augmented generation (RAG) (Du et al., 2025; Xu et al., 2023; Chen et al., 2023; Yoon et al., 2024). There have been a few summarization methods focused on prompt (rather than context) minimization, including Style-Compress (Pu et al., 2024), ATF (Jiang et al., 2025), and Nano-Capsulator (Chuang et al., 2024). Though these methods have shown success, they are limited —Style-Compress is deliberately task-specific; ATF is designed to remove noisy information rather than minimize relevant information; and Nano-Capsulator requires extensive training. In this study, we propose three task-agnostic prompt minimization techniques for no/low-training prompts. To our best knowledge, no prior work has directly addressed this question; thus, this project aims to bridge that gap.

## 3 Approach

We propose three approaches for task-agnostic prompt minimization. These approaches follow the same general framework, with slight variations in how minimization is approached. Notably, all three of these approaches are task-agnostic and require relatively little training.

### 3.1 Zero-Shot Learning

First, as a baseline, we propose a zero-shot LLM-based minimization that consists of two steps: minimization and evaluation. See Figure 1. Given an Original Prompt, we ask an

LLM to condense the prompt as much as possible. Then, the new prompt and its output are compared to the original prompt and output. Both stages are connected in a langchain pipeline that automates the minimization and evaluation process.

During stage (1), the minimization phase, the LLM is fed the original prompt and instructed to find the minimal possible candidate prompt that still preserves its core meaning. This approach is considered zero-shot because the only information provided is the original prompt, and the LLM is a system prompt engineered to complete the task in a single step with no reasoning.

After a candidate prompt is generated, stage (2) evaluates the prompt compared to the original prompt. First, the candidate prompt is presented to a chat-completion LLM, which generates a new output. Then, the candidate prompt and its corresponding output are compared to the original prompt and output. The comparison considers both the compression ratio of the prompts (by length) and the BERT similarity of the outputs.

Using the generated candidate prompt as input to stage one, it is possible to find an even smaller representation. Because the entire process is automated, the approach can be iterated until a desired stopping criterion is met.

### 3.2 In-Context Learning

Our second approach extends the zero-shot framework by formulating prompt minimization as a discrete optimization problem solved via an evolutionary strategy. Unlike the zero-shot approach, which generates a single trajectory of reductions, this method maintains a frontier of candidate minimal prompts to avoid local minima. We employ an LLM using In-Context Learning (ICL) to generate diverse prompt candidates. Figure 2 illustrates the two-stage optimization process across  $T$  iterations.

For stage (1), we generate a new generation of candidate prompts. For the initial iteration, we derive candidates directly from the original prompt. For future iterations, we seed the prompts by sampling from the historical population of top prompts. To balance exploration (diversity) and exploitation (quality), we em-

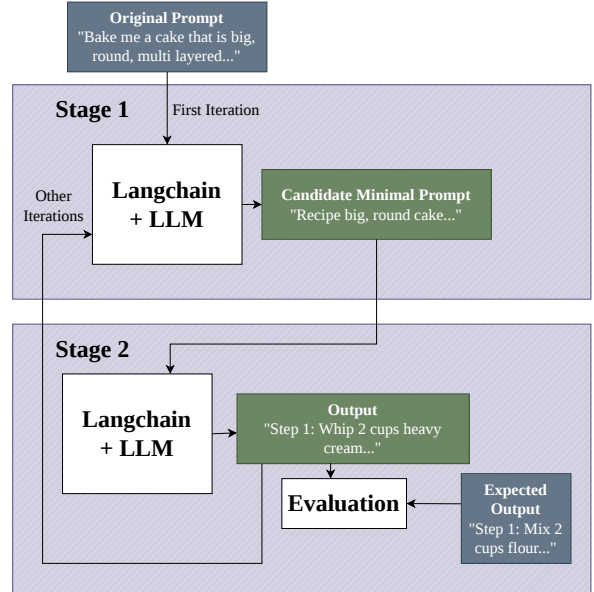


Figure 1: Flowchart illustrating proposed approach 1 (Zero-Shot Learning). We use an LLM to minimize a long prompt and then compare that minimized prompt’s output to the expected output. This variant generates one candidate minimal prompt and uses zero-shot learning.

ploy a weighted sampling selection mechanism based on inverse-fitness weighting,

$$P(p_i) \propto \frac{1}{\text{Score}(p_i) + 1},$$

such that a lower score implies a better minimal prompt.

These seed parent prompts undergo a mutation process. Similar to the zero-shot approach, the LLM is fed a system prompt containing potential insights for valid compressions (e.g., removing politeness markers, condensing instructions) to guide it in compressing the selected parent prompt while retaining semantic intent.

Stage (2) evaluates the newly generated candidate prompts by passing them to the same LLM to generate the outputs. These outputs are scored in the same way as the zero-shot approach, using a weighted objective function that combines the compression ratio and BERTScore semantic similarity. We then apply truncation selection (elitism) at the end of each iteration: only the global top- $N$  prompts are retained in the priority queue to serve as potential seeds for the next iteration. This cycle continues until the maximum iteration count is reached or convergence is detected.

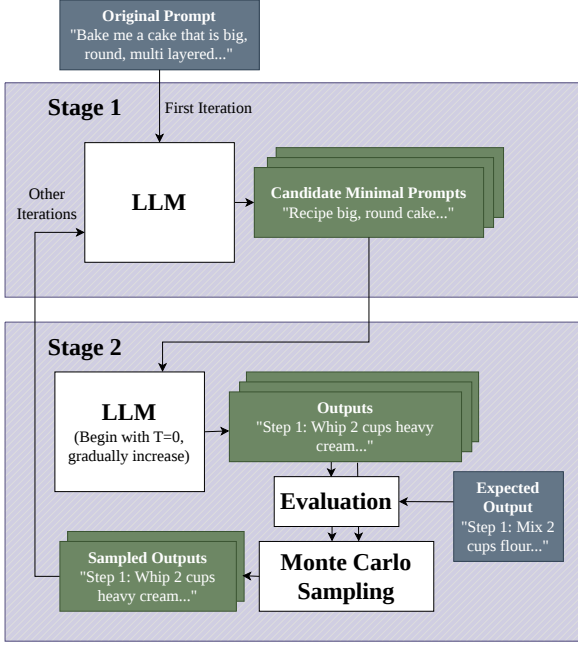


Figure 2: Flowchart illustrating proposed approach 2 (In-Context Learning). We use an LLM to minimize a long prompt, then compare the resulting prompt’s output to the expected output. This variant generates multiple candidate minimal prompts and uses in-context learning.

### 3.3 RL-Fine Tuning

Lastly, we propose a reinforcement learning (RL)- based framework to train an automated prompt compressor that learns a robust rewriting policy. This approach uses Proximal Policy Optimization (PPO) to fine-tune an LLM for semantic compression. To maintain computational efficiency and preserve the model’s general reasoning and language capabilities, we freeze the pre-trained backbone and optimize only a set of low-rank weights (LoRA). Figure 3 illustrates this training pipeline.

Our training process consists of two stages: (1) Policy Rollout and (2) Reward Calculation and Optimization.

In stage (1), the model, acting as the policy, generates a compressed version of the prompt. Unlike standard decoding, we inject dynamic systems instructions that evolve based on previous performance. For example, if the policy fails to compress or loses semantic meaning in a prior step sufficiently, explicit constraints are added to the context window. This mimics a “chain of thought” correction process, guiding the exploration through the solution space.

In stage (2), we evaluate the generated can-

didate using a composite reward function that balances semantic fidelity and token reduction. We use BERTScore to measure the embedding alignment between the original and generated outputs, ensuring instruction adherence while simultaneously calculating a compression ratio score. These signals are compressed into a scalar reward, which is used to compute the advantage and to update the LoRA adapters via PPO. This allows the model to gradually converge on a policy that aggressively removes redundancy while strictly maintaining the original prompt’s intent.

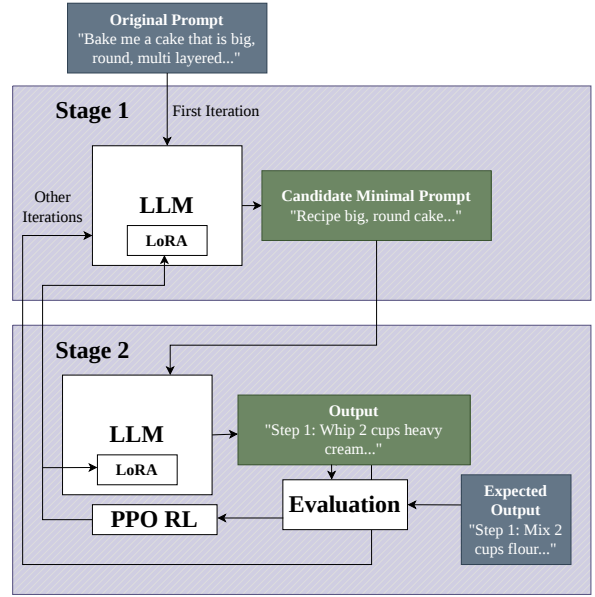


Figure 3: Flowchart illustrating proposed approach 3 (RL-Fine Tuning). We use an LLM to minimize a long prompt and then compare that minimized prompt’s output to the expected output. This variant fine-tunes the LLM using LoRA and PPO RL.

## 4 Experiments

### 4.1 Dataset Curation

Due to the limited availability of long, open-ended natural-language question prompts, we used OpenAI’s GPT-4o and GPT-5.1, Grok, and Claude AI’s Sonnet 4.6 to generate 60 long, open-ended prompts. The prompts ranged from 60 to 232 words, with an average of 128. These prompts span a wide range of domains, such as mathematical problem-solving, interpersonal relationship challenges, creative writing, and more.

The system prompt used to generate the dataset was:

**Data Generation Prompt**

*Can you generate 10 example prompts that I could use for a prompt-minimization dataset? The ideal prompt to minimize would be an open-ended question with context, so that it could be minimized. Make the prompts span a wide range of topics, such as math, law, engineering, creative writing, finance, and more. The prompt should be at least 250 words. Provide just the initial prompt.*

The full dataset of prompts is available for viewing and downloading in the code repository.

## 4.2 Evaluation Metrics

Our evaluation focuses on two key factors: semantic similarity and length compression.

We denote  $\mathbf{y}_0$  as the initial prompt’s generated output and  $\mathbf{y}_i$  as the generated output for the  $i$ th prompt.

### 4.2.1 Semantic similarity

We used BERTScore (Zhang et al., 2020) to measure semantic similarity.

In our experiment, we use this to evaluate whether the generated outputs hold the same meaning, regardless of specific sentence structure or word choice.

### 4.2.2 Compression

We determine length compression as the ratio of the lengths of the compressed prompt and the original prompt, or:

$$\text{Compression Score}(\mathbf{y}_i) = \frac{|\mathbf{y}_i|}{|\mathbf{y}_0|}$$

where  $|\cdot|$  denotes the length (in characters) of a string.

In our experiment, we use this to evaluate the compression level of a candidate minimal prompt.

## 5 Conclusions

This work studied prompt minimization: finding short, information-dense natural language prompts that preserve a model’s output behavior. We presented three complementary procedures—zero-shot search, in-context evolutionary search, and an RL-flavored variant

—and evaluated them using a two-term objective that trades off semantic fidelity and compression. Across diverse, long-form instructions, the procedures reliably discovered substantially shorter prompts whose outputs remained close in meaning to those of the originals. With the baseline explicitly set to iteration 0 and a baseline score of 0.5, we observe a sharp improvement in the next one to three iterations, followed by diminishing returns; the trajectory visualizations make these phases evident (Figs. 4, 5). Taken together, our results suggest that a significant fraction of typical prompts is redundant and that simple, training-light search already recovers strong compressions.

### 5.1 Takeaways

Two themes recur across models and methods. First, most gains arrive early: after the baseline (iteration 0), total loss typically drops steeply within a few steps before plateauing, as reflected in Figs. 4 and 5. Second,

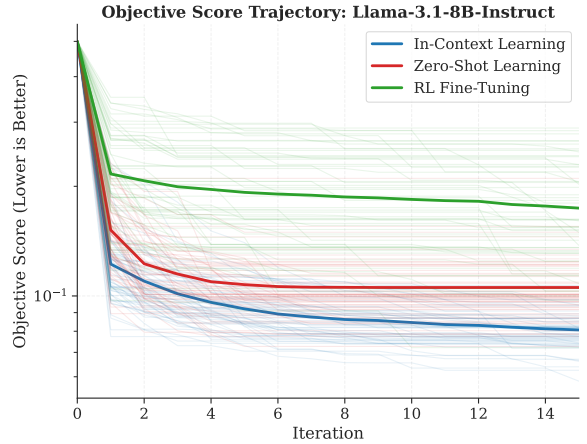


Figure 4: Best score trajectory of the 60 long context prompts using the meta-llama/Llama-3.1-8B-Instruct LLM. BERTScore and compression score are weighted equally at 0.5.

compression behavior is model-dependent: under matched settings, Llama-3.1-8B achieves stronger compression than Qwen-2.5-32B at a similar BERTScore (Table 1). This suggests that smaller models may compress more aggressively while still preserving semantic similarity. Methodologically, keeping a small frontier and resampling around it produced more stable progress than pure zero-shot proposals. Finally, the fixed 0.5/0.5 weighting between

| Algorithm           | Llama 3.1 8B                      |                 | Qwen2.5 32B                       |                 |
|---------------------|-----------------------------------|-----------------|-----------------------------------|-----------------|
|                     | Comp                              | BERT            | Comp                              | BERT            |
| Zero-Shot Learning  | $0.08 \pm 0.05$                   | $0.88 \pm 0.03$ | <b><math>0.17 \pm 0.08</math></b> | $0.89 \pm 0.02$ |
| In-Context Learning | <b><math>0.04 \pm 0.02</math></b> | $0.89 \pm 0.02$ | $0.19 \pm 0.10$                   | $0.90 \pm 0.01$ |
| RL Fine-Tuning      | $0.22 \pm 0.11$                   | $0.89 \pm 0.02$ | $0.25 \pm 0.09$                   | $0.89 \pm 0.02$ |

Table 1: Aggregate statistics for the different algorithms using the different LLMs

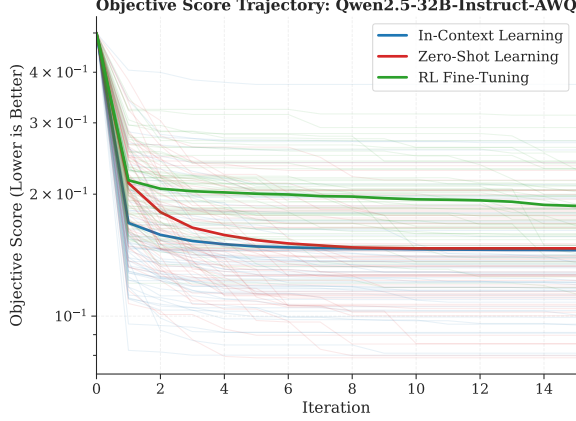


Figure 5: Best score trajectory of the 60 long context prompts using the Qwen/Qwen2.5-32B-Instruct-AWQ LLM. BERTScore and compression score are weighted equally at 0.5.

compression and similarity clearly shaped the discovered frontier: heavier emphasis on similarity preserves structure, whereas heavier emphasis on compression would push toward terser prompts at the risk of meaning drift. Moreover, the cross-model scatter (as reflected in Figs. 6 and 7) shows Llama-3.1-8B reaches lower compression ratios while keeping BERTScore effectively on par with Qwen-2.5-32B, i.e., stronger compression at comparable semantics.

## 5.2 Limitations

Iterative search with multiple candidates per step can be expensive in wall-clock time and tokens, especially when temperatures or frontier sizes are swept. Our fidelity estimates rely on embedding/BERT-style similarity, which may miss pragmatic nuances and can over- or under-penalize paraphrases. Several of the long prompts were machine-generated; outcomes may differ for expert-authored prompts with strict correctness constraints. Finally, a minimal prompt that reproduces a single reference output may not generalize to small task perturbations or broader input distributions.

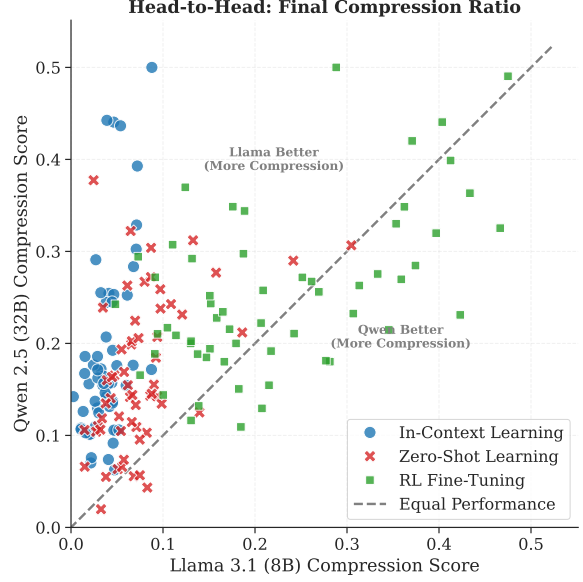


Figure 6: Best compression score compared between Qwen/Qwen2.5-32B-Instruct-AWQ and meta-llama/Llama-3.1-8B-Instruct for the same prompt. Lower is better.

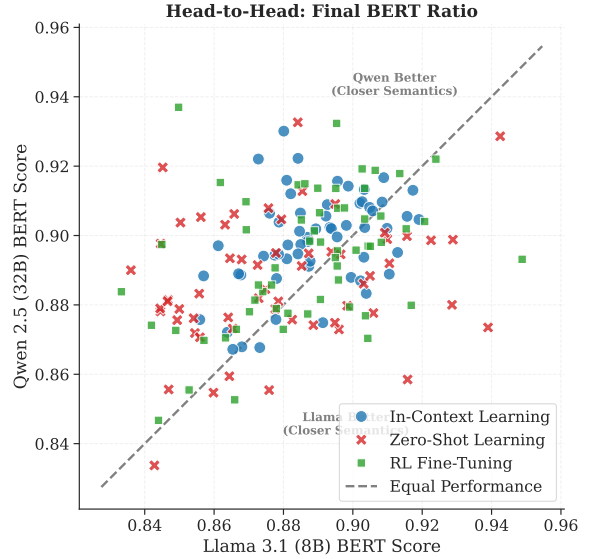


Figure 7: Best BERT score compared between Qwen/Qwen2.5-32B-Instruct-AWQ and meta-llama/Llama-3.1-8B-Instruct for the same prompt. Lower is better.

### 5.3 Future Work

A natural next step is to complement automatic similarity with human preference studies or lightweight preference models and to add an explicit quality/helpfulness term to the objective. Rather than a single weighted sum, we aim to expose a Pareto frontier over similarity, compression, cost, and safety so practitioners can choose operating points. Richer fidelity signals, such as NLI consistency, factuality checks, or task-specific scorers, could reduce metric gaming. Cost can be reduced with adaptive early stopping and per-prompt candidate budgets learned from online improvement rates. Finally, conditioning the minimizer on task type and model family, extending minimization to retrieved contexts and system prompts with safety filters, and packaging the visualizer and scripts into a small library with a short practitioner handout should make these techniques easy to adopt in practice.

### 5.4 Code availability

The code is available at the GitHub repo [jontgao/546-prompt-minimization](https://github.com/jontgao/546-prompt-minimization)

The output logs and runs are available at the HuggingFace repository [SuperComputer/PromptMinimization](https://huggingface.co/SuperComputer/PromptMinimization)

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## A Result examples

**Zero-Shot Learning Examples:** Figures 8 and 9 use the Algorithm in 3.1 for generating the minimization of the prompts.

**In-Context Learning Examples:** Figures 10 and 11 use the Algorithm in 3.2 for generating the minimization of the prompts.

**RL Fine-Tuning Examples:** Figures 12 and 13 use the Algorithm in 3.3 for generating the minimization of the prompts.

### Zero-Shot Learning: Initial Prompt

*As artificial intelligence systems continue integrating into nearly every aspect of daily life—from personalized assistants that anticipate our needs to automated systems that influence hiring, finance, healthcare, and public policy—the question of how humans and machines should coexist becomes increasingly complex. Beyond simply determining when machines should take over tasks, society must grapple with how AI reshapes human agency, autonomy, and social structures. Considering the tradeoffs between convenience, efficiency, and control, how do you envision the ideal balance between human judgment and machine autonomy? What cultural shifts, safeguards, regulatory frameworks, or ethical principles do you believe are necessary to ensure that these technologies enhance human well-being while protecting individual freedoms, preventing algorithmic biases, and preserving meaningful human oversight?*

#### Qwen/Qwen2.5-32B-Instruct-AWQ

**Iter 1** (Score: 0.1648 | Bert: 0.8806 | Comp: 0.2102)

How can we balance human judgment and AI autonomy to enhance well-being while preventing biases and ensuring oversight? Discuss necessary cultural shifts, safeguards, and ethical principles.

**Iter 2** (Score: 0.1337 | Bert: 0.8864 | Comp: 0.1538)

Balance human judgment and AI to enhance well-being, prevent biases, and ensure oversight. Discuss cultural shifts, safeguards, and ethics.

**Iter 3** (Score: 0.1318 | Bert: 0.8912 | Comp: 0.1549)

Balance human judgment with AI to enhance well-being, prevent biases, and ensure oversight. Discuss cultural shifts, safeguards, and ethics.

**Iter 4** (Score: 0.1257 | Bert: 0.8892 | Comp: 0.1405)

Balance human judgment with AI for well-being, bias prevention, and oversight. Discuss cultural shifts, safeguards, and ethics.

#### meta-llama/Llama-3.1-8B-Instruct

**Iter 1** (Score: 0.1299 | Bert: 0.8851 | Comp: 0.1449)

How can humans and AI systems balance autonomy and agency in a way that enhances human well-being and protects individual freedoms?

**Iter 2** (Score: 0.1055 | Bert: 0.8852 | Comp: 0.0962)

Balancing AI autonomy and human agency for enhanced well-being and individual freedoms.

**Iter 3** (Score: 0.0943 | Bert: 0.8844 | Comp: 0.0730)

Balancing AI autonomy and human agency for well-being and freedom.

**Iter 4** (Score: 0.0803 | Bert: 0.8826 | Comp: 0.0431)

Balancing AI autonomy and human agency.

Figure 8: **Milestone Discoveries in Prompt Minimization using Zero-Shot Learning.** The top panel shows the verbose initial prompt. The bottom panels compare the minimization trajectory of Qwen-32B (Left) and Llama-3.1-8B (Right) using Algorithm from Section 3.1. Metrics indicate total score, BERT-score, and compression ratio. Running with Temperature 0.

| Zero-Shot Learning: Initial Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <p><i>Explain the process of photosynthesis to 10th-grade biology students, including the major steps like light absorption by chlorophyll, the splitting of water molecules to release oxygen, and the Calvin cycle for sugar production. Make sure to cover how energy from the sun is converted into chemical energy stored in glucose, the role of stomata in gas exchange, and real-world examples of how disruptions like deforestation affect this process on a global scale, while keeping the language accessible and avoiding overly technical jargon.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Qwen/Qwen2.5-32B-Instruct-AWQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | meta-llama/Llama-3.1-8B-Instruct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Iter 1</b> (Score: 0.2466   Bert: 0.9026   Comp: 0.3959)<br/>           Explain photosynthesis to 10th graders: light absorption, water splitting for oxygen, Calvin cycle for sugar. Show sun energy to glucose, stomata for gas exchange, and impact of deforestation. Use simple language.</p> <p><b>Iter 2</b> (Score: 0.2206   Bert: 0.9286   Comp: 0.3699)<br/>           Explain photosynthesis to 10th graders: light absorption, water splitting for oxygen, Calvin cycle for sugar. Show sun to glucose, stomata for gas exchange, deforestation impact. Use simple language.</p> <p><b>Iter 3</b> (Score: 0.2187   Bert: 0.8917   Comp: 0.3290)<br/>           Explain photosynthesis to 10th graders: light, water to oxygen, Calvin cycle for sugar. Show sun to glucose, stomata for gas exchange, deforestation impact. Use simple language.</p> <p><b>Iter 4</b> (Score: 0.2097   Bert: 0.8780   Comp: 0.2974)<br/>           Explain photosynthesis to 10th graders: light, water to oxygen, Calvin cycle for sugar. Show sun to glucose, stomata, deforestation impact. Use simple language.</p> <p><b>Iter 5</b> (Score: 0.1944   Bert: 0.9012   Comp: 0.2900)<br/>           Explain photosynthesis to 10th graders: light, water to oxygen, Calvin cycle for sugar. Show sun to glucose, stomata, deforestation impact. Simple language.</p> | <p><b>Iter 1</b> (Score: 0.2027   Bert: 0.9310   Comp: 0.3364)<br/>           Explain photosynthesis to 10th-grade biology students, covering light absorption, water splitting, the Calvin cycle, energy conversion, stomata, and global impacts of deforestation.</p> <p><b>Iter 2</b> (Score: 0.1854   Bert: 0.9173   Comp: 0.2881)<br/>           Explain photosynthesis to 10th-grade biology students, covering light absorption, the Calvin cycle, energy conversion, and global impacts of deforestation.</p> <p><b>Iter 3</b> (Score: 0.1673   Bert: 0.9349   Comp: 0.2695)<br/>           Explain photosynthesis to 10th-grade biology students, covering light absorption, the Calvin cycle, energy conversion, and deforestation impacts.</p> <p><b>Iter 4</b> (Score: 0.1673   Bert: 0.9350   Comp: 0.2695)<br/>           Explain photosynthesis to 10th-grade biology students, covering light absorption, the Calvin cycle, energy conversion, and deforestation effects.</p> <p><b>Iter 5</b> (Score: 0.1496   Bert: 0.9425   Comp: 0.2416)<br/>           Explain photosynthesis to 10th graders, covering light absorption, the Calvin cycle, energy conversion, and deforestation impacts.</p> |

Figure 9: **Milestone Discoveries in Prompt Minimization using Zero-Shot Learning.** The top panel shows the verbose initial prompt. The bottom panels compare the minimization trajectory of Qwen-32B (Left) and Llama-3.1-8B (Right) using Algorithm from Section 3.1. Metrics indicate total score, BERT-score, and compression ratio. Running with Temperature 0.

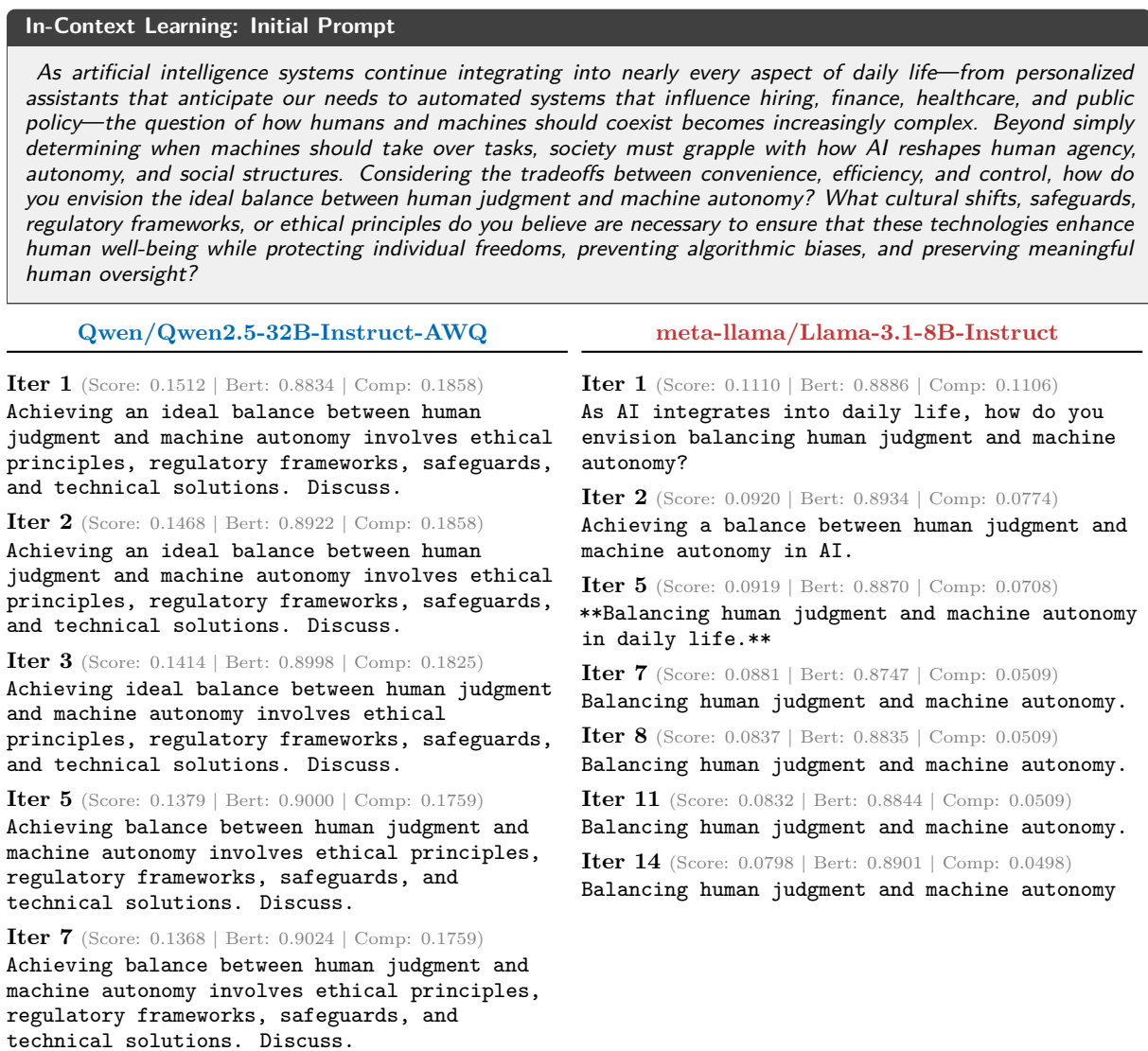


Figure 10: **Milestone Discoveries in Prompt Minimization using In-Context Learning.** The top panel shows the verbose initial prompt. The bottom panels compare the minimization trajectory of Qwen-32B (Left) and Llama-3.1-8B (Right) using Algorithm from Section 3.2. Metrics indicate total score, BERT-score, and compression ratio. Running with Temperature 0.9.

| In-Context Learning: Initial Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <p><i>Explain the process of photosynthesis to 10th-grade biology students, including the major steps like light absorption by chlorophyll, the splitting of water molecules to release oxygen, and the Calvin cycle for sugar production. Make sure to cover how energy from the sun is converted into chemical energy stored in glucose, the role of stomata in gas exchange, and real-world examples of how disruptions like deforestation affect this process on a global scale, while keeping the language accessible and avoiding overly technical jargon.</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Qwen/Qwen2.5-32B-Instruct-AWQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | meta-llama/Llama-3.1-8B-Instruct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><b>Iter 1</b> (Score: 0.2851   Bert: 0.9019   Comp: 0.4721)<br/>           Explain photosynthesis to 10th-grade students: light absorption by chlorophyll, water splitting to release oxygen, Calvin cycle for sugar. Cover energy conversion to glucose, role of stomata in gas exchange, impact of deforestation. Keep language simple.</p> <p><b>Iter 2</b> (Score: 0.2767   Bert: 0.8982   Comp: 0.4517)<br/>           Explain photosynthesis to 10th-grade students: light absorption by chlorophyll, water splitting to release oxygen, Calvin cycle for sugar. Cover energy conversion to glucose, stomata role, and impact of deforestation. Keep language accessible.</p> <p><b>Iter 3</b> (Score: 0.2743   Bert: 0.8919   Comp: 0.4405)<br/>           Explain photosynthesis to 10th-grade students: light absorption by chlorophyll, water splitting to release oxygen, Calvin cycle for sugar, energy conversion to glucose, stomata for gas exchange, impacts of deforestation. Keep accessible.</p> <p><b>Iter 4</b> (Score: 0.2724   Bert: 0.8958   Comp: 0.4405)<br/>           Explain photosynthesis to 10th-grade students: light absorption by chlorophyll, water splitting to release oxygen, Calvin cycle for sugar, energy conversion to glucose, stomata for gas exchange, impacts of deforestation. Keep accessible.</p> <p><b>Iter 6</b> (Score: 0.2719   Bert: 0.8967   Comp: 0.4405)<br/>           Explain photosynthesis to 10th-grade students: light absorption by chlorophyll, water splitting to release oxygen, Calvin cycle for sugar, energy conversion to glucose, stomata for gas exchange, impacts of deforestation. Keep accessible.</p> <p><b>Iter 7</b> (Score: 0.2713   Bert: 0.8978   Comp: 0.4405)<br/>           Explain photosynthesis to 10th-grade students: light absorption by chlorophyll, water splitting to release oxygen, Calvin cycle for sugar, energy conversion to glucose, stomata for gas exchange, impacts of deforestation. Keep accessible.</p> <p><b>Iter 8</b> (Score: 0.2705   Bert: 0.8995   Comp: 0.4405)<br/>           Explain photosynthesis to 10th-grade students: light absorption by chlorophyll, water splitting to release oxygen, Calvin cycle for sugar, energy conversion to glucose, stomata for gas exchange, impacts of deforestation. Keep accessible.</p> | <p><b>Iter 1</b> (Score: 0.1817   Bert: 0.8819   Comp: 0.2454)<br/> <b>**Step-by-step explanation of photosynthesis for 10th-grade biology students: 5 steps, accessible language, real-world disruptions**</b></p> <p><b>Iter 2</b> (Score: 0.1532   Bert: 0.8795   Comp: 0.1859)<br/> <b>**Step-by-step explanation of photosynthesis: 5 steps, accessible language, real-world disruptions**</b></p> <p><b>Iter 3</b> (Score: 0.1403   Bert: 0.8717   Comp: 0.1524)<br/> <b>**5-step photosynthesis explanation: accessible language, real-world disruptions**</b></p> <p><b>Iter 4</b> (Score: 0.1286   Bert: 0.8711   Comp: 0.1283)<br/>           5-step photosynthesis explanation, 10th-grade, real-world disruptions</p> <p><b>Iter 5</b> (Score: 0.1212   Bert: 0.8766   Comp: 0.1190)<br/> <b>**5-step explanation of photosynthesis, real-world disruptions**</b></p> <p><b>Iter 6</b> (Score: 0.1089   Bert: 0.8733   Comp: 0.0911)<br/>           Photosynthesis in 5 steps, real-world disruptions</p> <p><b>Iter 8</b> (Score: 0.1036   Bert: 0.8765   Comp: 0.0836)<br/> <b>**Photosynthesis in 5 steps and disruptions**</b></p> <p><b>Iter 9</b> (Score: 0.1016   Bert: 0.8804   Comp: 0.0836)<br/> <b>**Photosynthesis in 5 steps and disruptions**</b></p> <p><b>Iter 10</b> (Score: 0.0977   Bert: 0.8882   Comp: 0.0836)<br/> <b>**Photosynthesis in 5 steps and disruptions**</b></p> <p><b>Iter 12</b> (Score: 0.0907   Bert: 0.8818   Comp: 0.0632)<br/>           Photosynthesis 5 steps, accessible</p> <p><b>Iter 13</b> (Score: 0.0841   Bert: 0.8801   Comp: 0.0483)<br/>           Photosynthesis in 5 steps.</p> <p><b>Iter 14</b> (Score: 0.0786   Bert: 0.8893   Comp: 0.0465)<br/>           Photosynthesis in 5 steps</p> |

Figure 11: Milestone Discoveries in Prompt Minimization using In-Context Learning. The top panel shows the verbose initial prompt. The bottom panels compare the minimization trajectory of Qwen-32B (Left) and Llama-3.1-8B (Right) using Algorithm from Section 3.2. Metrics indicate total score, BERT-score, and compression ratio. Running with Temperature 0.9.

| RL Fine-Tuning: Initial Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p><i>As artificial intelligence systems continue integrating into nearly every aspect of daily life—from personalized assistants that anticipate our needs to automated systems that influence hiring, finance, healthcare, and public policy—the question of how humans and machines should coexist becomes increasingly complex. Beyond simply determining when machines should take over tasks, society must grapple with how AI reshapes human agency, autonomy, and social structures. Considering the tradeoffs between convenience, efficiency, and control, how do you envision the ideal balance between human judgment and machine autonomy? What cultural shifts, safeguards, regulatory frameworks, or ethical principles do you believe are necessary to ensure that these technologies enhance human well-being while protecting individual freedoms, preventing algorithmic biases, and preserving meaningful human oversight?</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Qwen/Qwen2.5-32B-Instruct-AWQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | meta-llama/Llama-3.1-8B-Instruct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Iter 1</b> (Score: 0.2293   Bert: 0.8744   Comp: 0.3330)<br/> Analyze AI's impact on human life in areas like assistants, hiring, finance, healthcare, &amp; policy. Evaluate balance btwn human judgment &amp; machine autonomy. Identify nec. cultural shifts, safeguards, regulations, &amp; ethics to enhance well-being, protect freedoms, prevent bias, &amp; ensure human oversight.</p> <p><b>Iter 2</b> (Score: 0.2257   Bert: 0.8926   Comp: 0.3440)<br/> AI integrates into daily life (assistants, hiring, finance, healthcare, policy). Define ideal balance btwn human judgment &amp; machine autonomy. Consider conven, effi, ctrl. Suggest cultural shifts, safeguards, reg frameworks, ethics to enhance well-being, protect freedoms, prevent biases, ensure human oversight.</p> <p><b>Iter 3</b> (Score: 0.2217   Bert: 0.8620   Comp: 0.3053)<br/> Analyze AI integration impact on human life, task delegation, agency, &amp; social struct. Discuss balance btw convenience, efficiency, control. Suggest nec. cultural shifts, safeguards, regs, ethics to enhance well-being, protect freedoms, prevent biases, ensure human oversight.</p> <p><b>Iter 5</b> (Score: 0.2092   Bert: 0.8891   Comp: 0.3075)<br/> AI integrates into daily life (assistants, hiring, finance, healthcare, policy). Discuss balance between human judgment &amp; machine autonomy. Outline cultural shifts, safeguards, regulations, ethics needed to enhance well-being, protect freedoms, prevent biases, ensure oversight.</p> <p><b>Iter 13</b> (Score: 0.1979   Bert: 0.8984   Comp: 0.2942)<br/> AI integrates into daily life (assistants, hiring, finance, healthcare, policy). Define balance btwn human judgment &amp; machine autonomy. Identify nec. cultural shifts, safeguards, regs, ethics to enhance well-being, protect freedoms, prevent biases, ensure oversight.</p> | <p><b>Iter 1</b> (Score: 0.1178   Bert: 0.8806   Comp: 0.1162)<br/> &gt; Determine the ideal balance between human judgment &amp; machine autonomy as AI integrates into daily life.</p> <p><b>Iter 7</b> (Score: 0.1166   Bert: 0.8852   Comp: 0.1184)<br/> Determine the ideal balance between human judgment &amp; machine autonomy in an increasingly AI-driven society.</p> <p><b>Iter 10</b> (Score: 0.1110   Bert: 0.8876   Comp: 0.1095)<br/> &gt; Determine the ideal balance between human judgment &amp; machine autonomy in an AI-dominated society.</p> <p><b>Iter 11</b> (Score: 0.1014   Bert: 0.8746   Comp: 0.0774)<br/> Determine the ideal balance between human judgment &amp; machine autonomy.</p> <p><b>Iter 14</b> (Score: 0.0973   Bert: 0.8783   Comp: 0.0730)<br/> &gt; Determine the balance between human judgment &amp; machine autonomy.</p> |

Figure 12: **Milestone Discoveries in Prompt Minimization using RL Fine-Tuning.** The top panel shows the verbose initial prompt. The bottom panels compare the minimization trajectory of Qwen-32B (Left) and Llama-3.1-8B (Right) using Algorithm from Section 3.3. Metrics indicate total score, BERT-score, and compression ratio. Running with Temperature 0.8.

| RL Fine-Tuning: Initial Prompt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <p><i>Explain the process of photosynthesis to 10th-grade biology students, including the major steps like light absorption by chlorophyll, the splitting of water molecules to release oxygen, and the Calvin cycle for sugar production. Make sure to cover how energy from the sun is converted into chemical energy stored in glucose, the role of stomata in gas exchange, and real-world examples of how disruptions like deforestation affect this process on a global scale, while keeping the language accessible and avoiding overly technical jargon.</i></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Qwen/Qwen2.5-32B-Instruct-AWQ                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | meta-llama/Llama-3.1-8B-Instruct                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>Iter 1</b> (Score: 0.2161   Bert: 0.8931   Comp: 0.3253)</p> <p>Explain photosynthesis to 10th graders: light 吸收 by 叶绿素, 水分子 split to release O<sub>2</sub>, Calvin cycle for 糖 production. Cover 阳光转化 学能 in 葡萄糖, 气孔 role in 气体交换, &amp; how 破坏如伐木影响全球光合作用. Use accessible 语言.</p>                                                                                                                                                                                                                                                                                 | <p><b>Iter 1</b> (Score: 0.2750   Bert: 0.9165   Comp: 0.4665)</p> <p>Explain photosynthesis to 10th-grade biology students, covering light absorption by chlorophyll, water molecule splitting, oxygen release, Calvin cycle for sugar production, energy conversion, stomata function, &amp; global disruptions like deforestation.</p> <p><b>Iter 3</b> (Score: 0.2663   Bert: 0.9488   Comp: 0.4814)</p> <p>Explain photosynthesis to 10th-grade biology students, covering light absorption by chlorophyll, water molecule splitting &amp; oxygen release, the Calvin cycle, energy conversion to glucose, stomata's role in gas exchange &amp; global disruptions like deforestation.</p> |

Figure 13: **Milestone Discoveries in Prompt Minimization using RL Fine-Tuning.** The top panel shows the verbose initial prompt. The bottom panels compare the minimization trajectory of Qwen-32B (Left) and Llama-3.1-8B (Right) using Algorithm from Section 3.3. Metrics indicate total score, BERT-score, and compression ratio. Running with Temperature 0.8.